HIGH PERFORMANCE COMPUTING

Assginment-1

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Section 1: Section A — SSH Access & Environment Sanity Check (Login Node)

```
%%bash python --
version
```

```
Python 3.12.12
```

```
%%bash which
python which
python3
```

```
/usr/local/bin/python
/usr/bin/python3
```

```
%%bash lscpu
```

```
Architecture:                         x86_64
CPU op-mode(s):                       32-bit, 64-bit
Address sizes:                        46 bits physical, 48 bits virtual
Byte Order:                           Little Endian
CPU(s):                               2
On-line CPU(s) list:                  0,1
Vendor ID:                            GenuineIntel
Model name:                           Intel(R) Xeon(R) CPU @ 2.20GHz
CPU family:                           6
Model:                                79
Thread(s) per core:                   2
Core(s) per socket:                   1
Socket(s):                            1
Stepping:                             0
BogoMIPS:                             4399.99
Flags:                                fpu vme de pse tsc msr pae mce cx8
apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall
nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology nonstop_tsc cpuid
tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe
popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd
ibrs ibpb stibp fsgsbase tsc_adjust bmi1 hle avx2 smep bmi2 erms invpcid rtm
rdseed adx smap xsaveopt arat md_clear arch_capabilities
Hypervisor vendor:                    KVM
Virtualization type:                  full
```

```
L1d cache:                               32 KiB (1 instance)
L1i cache:                               32 KiB (1 instance)
L2 cache:                                256 KiB (1 instance)
L3 cache:                                55 MiB (1 instance)
NUMA node(s):                            1
NUMA node0 CPU(s):                       0,1
Vulnerability Gather data sampling:      Not affected
Vulnerability Indirect target selection: Vulnerable
Vulnerability Itlb multihit:             Not affected
Vulnerability L1tf:                      Mitigation; PTE Inversion
Vulnerability Mds:                       Vulnerable; SMT Host state unknown
Vulnerability Meltdown:                  Vulnerable
Vulnerability Mmio stale data:           Vulnerable
Vulnerability Reg file data sampling:    Not affected
Vulnerability Retbleed:                  Vulnerable
Vulnerability Spec rstack overflow:      Not affected
Vulnerability Spec store bypass:         Vulnerable
Vulnerability Spectre v1:                Vulnerable: __user pointer
sanitization and usercopy barriers only; no swapgs barriers
Vulnerability Spectre v2:                Vulnerable; IBPB: disabled; STIBP:
disabled; PBRSB-eIBRS: Not affected; BHI: Vulnerable Vulnerability
Srbds:                      Not affected
Vulnerability Tsa:                       Not affected
Vulnerability Tsx async abort:           Vulnerable
```

```
%%bash cat
/proc/cpuinfo
```

```
processor : 0 vendor_id :
GenuineIntel cpu family :
6
model       : 79
model name : Intel(R) Xeon(R) CPU @ 2.20GHz
stepping : 0
microcode : 0xffffffff
cpu MHz           : 2199.998
cache size : 56320 KB
physical id : 0 siblings : 2
core id           : 0
cpu cores : 1
apicid            : 0
initial apicid : 0
fpu         : yes
fpu_exception     : yes cpuid
level : 13
wp          : yes
```

```
flags       : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov
pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm
constant_tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni
pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx
f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp
fsgsbase tsc_adjust bmi1 hle avx2 smep bmi2 erms invpcid rtm rdseed adx smap
xsaveopt arat md_clear arch_capabilities
bugs  : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs
taa mmio_stale_data retbleed bhi its
bogomips : 4399.99 clflush size     : 64
cache_alignment : 64 address sizes  : 46 bits
physical, 48 bits virtual power management:

processor : 1 vendor_id :
GenuineIntel cpu family :
6
model       : 79
model name : Intel(R) Xeon(R) CPU @ 2.20GHz
stepping : 0 microcode :
0xffffffff
cpu MHz           : 2199.998
cache size : 56320 KB
physical id : 0 siblings
: 2
core id           : 0
cpu cores : 1
apicid            : 1 initial
apicid : 1
fpu         : yes
fpu_exception     : yes cpuid
level : 13
wp          : yes
flags       : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov
pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm
constant_tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni
pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx
f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp
fsgsbase tsc_adjust bmi1 hle avx2 smep bmi2 erms invpcid rtm rdseed adx smap
xsaveopt arat md_clear arch_capabilities
bugs  : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs
taa mmio_stale_data retbleed bhi its
bogomips : 4399.99 clflush size     : 64
cache_alignment : 64 address sizes  : 46 bits
physical, 48 bits virtual power management:
```

```
%%bash nproc
2
```

```
%%bash
echo 'print("Hello from the HPC cluster!")' > sanity_check.py
cat sanity_check.py print("Hello from the HPC cluster!")

%%bash python
sanity_check.py

Hello from the HPC cluster!
```

Section 2: Section B — First Batch Job: Serial Python Script via Slurm/PBS

```
%%bash
cat <<EOF > submit_job.sh
#!/bin/bash
#SBATCH --job-name=MyPythonJob
#SBATCH --output=slurm-%j.out
#SBATCH --error=slurm-%j.err
#SBATCH --ntasks=1
#SBATCH --time=00:01:00

echo "Starting Python script..."
python sanity_check.py echo
"Python script finished." EOF

%%bash cat
submit_job.sh

#!/bin/bash
#SBATCH --job-name=MyPythonJob
#SBATCH --output=slurm-%j.out
#SBATCH --error=slurm-%j.err
#SBATCH --ntasks=1
#SBATCH --time=00:01:00

echo "Starting Python script..."
python sanity_check.py echo
"Python script finished."

%%bash bash submit_job.sh > slurm-12345.out 2> slurm-
12345.err

%%bash ls -l slurm-12345.out slurm-
12345.err

-rw-r--r-- 1 root root  0 Jan  6 04:49 slurm-12345.err
-rw-r--r-- 1 root root 78 Jan  6 04:49 slurm-12345.out
%%bash cat slurm-
12345.out
```

```
Starting Python script...
Hello from the HPC cluster! Python
script finished.
```

```
%%bash cat slurm-
12345.err
```

Section 3: Section C — Interactive Compute Session (salloc/srun or qsub -I)

```
%%bash
cat <<EOF > run_interactive_python.sh
#!/bin/bash
#SBATCH --job-name=InteractivePython
#SBATCH --output=interactive-%j.out
#SBATCH --error=interactive-%j.err
#SBATCH --ntasks=1
#SBATCH --time=00:01:00

echo "Launching simulated interactive session..."
python sanity_check.py echo "Simulated
interactive session finished."
EOF

# Run the script and redirect output to files bash
run_interactive_python.sh > interactive-12346.out 2>
interactive12346.err

echo "Script run_interactive_python.sh created and executed."
```

```
Script run_interactive_python.sh created and executed.
```

```
%%bash ls -l interactive-12346.out interactive-
12346.err
```

```
-rw-r--r-- 1 root root   0 Jan  6 05:03 interactive-12346.err
-rw-r--r-- 1 root root 111 Jan  6 05:03 interactive-12346.out
```

```
%%bash cat interactive-
12346.out
```

```
Launching simulated interactive session...
Hello from the HPC cluster!
Simulated interactive session finished.
```

```
%%bash cat interactive-
12346.err
```

Section 4: Section D — Job Arrays for Parameter Sweep (Multiple Serial Runs)

```
%%bash
```

```
cat <<EOF > run_monte_carlo.py
import sys import random

def estimate_pi(sample_size):
    inside_circle = 0    for _
in range(sample_size):        x
= random.uniform(0, 1)        y
= random.uniform(0, 1)
distance = x**2 + y**2
if distance <= 1:
            inside_circle += 1

    pi_estimate = 4 * inside_circle / sample_size
return pi_estimate

if __name__ == '__main__':    if len(sys.argv) != 2:
print("Usage: python run_monte_carlo.py <sample_size>")
sys.exit(1)

    sample_size = int(sys.argv[1])
pi_est = estimate_pi(sample_size)
    print(f"Sample Size: {sample_size}, Pi Estimate: {pi_est}")

EOF

cat run_monte_carlo.py

import sys
import random

def estimate_pi(sample_size):
    inside_circle = 0    for _
in range(sample_size):        x
= random.uniform(0, 1)        y
= random.uniform(0, 1)
distance = x**2 + y**2
if distance <= 1:
            inside_circle += 1

    pi_estimate = 4 * inside_circle / sample_size
return pi_estimate

if __name__ == '__main__':
if len(sys.argv) != 2:
        print("Usage: python run_monte_carlo.py <sample_size>")
sys.exit(1)
```

```
    sample_size = int(sys.argv[1])
pi_est = estimate_pi(sample_size)
    print(f"Sample Size: {sample_size}, Pi Estimate: {pi_est}")
```

Now, let's create the Slurm batch script `submit_array_job.sh`. This script will define an array job that runs `run_monte_carlo.py` with different sample sizes based on the `SLURM_ARRAY_TASK_ID` environment variable.

```
%%bash
cat <<EOF > submit_array_job.sh
#!/bin/bash
#SBATCH --job-name=MonteCarloArray
#SBATCH --output=array-job-%A_%a.out
#SBATCH --error=array-job-%A_%a.err
#SBATCH --ntasks=1
#SBATCH --time=00:02:00
#SBATCH --array=0-2 # This creates 3 tasks: 0, 1, 2

# Calculate sample size based on array task ID
# Task 0 -> 1000, Task 1 -> 2000, Task 2 -> 3000
SAMPLE_SIZE=$((1000 * (SLURM_ARRAY_TASK_ID + 1)))

echo "Starting Monte Carlo simulation for task $SLURM_ARRAY_TASK_ID with
sample size $SAMPLE_SIZE" python run_monte_carlo.py $SAMPLE_SIZE echo
"Finished Monte Carlo simulation for task $SLURM_ARRAY_TASK_ID" EOF

cat submit_array_job.sh
```

```
#!/bin/bash
#SBATCH --job-name=MonteCarloArray
#SBATCH --output=array-job-%A_%a.out
#SBATCH --error=array-job-%A_%a.err
#SBATCH --ntasks=1
#SBATCH --time=00:02:00
#SBATCH --array=0-2 # This creates 3 tasks: 0, 1, 2

# Calculate sample size based on array task ID
# Task 0 -> 1000, Task 1 -> 2000, Task 2 -> 3000
SAMPLE_SIZE=1000

echo "Starting Monte Carlo simulation for task  with sample size " python
run_monte_carlo.py
echo "Finished Monte Carlo simulation for task "
```

Since we are in a simulated environment, we will manually simulate the execution of the array job by looping through the array task IDs and running the script for each. We'll capture the output and error for each simulated task.

```
%%bash

JOB_ID=98765 # Simulate a job ID

for SLURM_ARRAY_TASK_ID in 0 1 2;
do
  SAMPLE_SIZE=$((1000 * (SLURM_ARRAY_TASK_ID + 1)))
  OUTPUT_FILE="array-job-${JOB_ID}_${SLURM_ARRAY_TASK_ID}.out"
  ERROR_FILE="array-job-${JOB_ID}_${SLURM_ARRAY_TASK_ID}.err"

  echo "Starting Monte Carlo simulation for task ${SLURM_ARRAY_TASK_ID} with
sample size ${SAMPLE_SIZE}" > "${OUTPUT_FILE}"
  python run_monte_carlo.py ${SAMPLE_SIZE} >> "${OUTPUT_FILE}" 2>>
"${ERROR_FILE}"   echo "Finished Monte Carlo simulation for task
${SLURM_ARRAY_TASK_ID}" >> "${OUTPUT_FILE}"

  echo "Simulated execution for task ${SLURM_ARRAY_TASK_ID}. Output in
${OUTPUT_FILE}, Error in ${ERROR_FILE}" done
```

```
Simulated execution for task 0. Output in array-job-98765_0.out, Error in
array-job-98765_0.err
Simulated execution for task 1. Output in array-job-98765_1.out, Error in
array-job-98765_1.err
Simulated execution for task 2. Output in array-job-98765_2.out, Error in
array-job-98765_2.err
```

Let's list the generated output and error files and then examine their contents.

```
%%bash ls -l array-job-*.out array-
job-*.err
```

```
-rw-r--r-- 1 root root   0 Jan  6 05:06 array-job-98765_0.err
-rw-r--r-- 1 root root 146 Jan  6 05:06 array-job-98765_0.out
-rw-r--r-- 1 root root   0 Jan  6 05:06 array-job-98765_1.err
-rw-r--r-- 1 root root 146 Jan  6 05:06 array-job-98765_1.out
-rw-r--r-- 1 root root   0 Jan  6 05:06 array-job-98765_2.err
-rw-r--r-- 1 root root 159 Jan  6 05:06 array-job-98765_2.out
```

```
%%bash
cat array-job-98765_0.out cat
array-job-98765_1.out cat
array-job-98765_2.out
```

```
Starting Monte Carlo simulation for task 0 with sample size 1000
Sample Size: 1000, Pi Estimate: 3.108
Finished Monte Carlo simulation for task 0
Starting Monte Carlo simulation for task 1 with sample size 2000
Sample Size: 2000, Pi Estimate: 3.086
```

```
Finished Monte Carlo simulation for task 1
Starting Monte Carlo simulation for task 2 with sample size 3000
Sample Size: 3000, Pi Estimate: 3.1226666666666665
Finished Monte Carlo simulation for task 2
```

```
%%bash cat array-job-
98765_0.err cat array-
job-98765_1.err cat
array-job-98765_2.err
```

Section 5: Section E — Modules/Venv, Scratch I/O, and Resource Flags

```
%%bash
cat <<EOF > scratch_io_script.py
import sys import os

def main():     if len(sys.argv) != 2:         print("Usage: python
scratch_io_script.py <scratch_directory_path>")         sys.exit(1)

    scratch_dir = sys.argv[1]

    # Create the scratch directory if it doesn't exist
os.makedirs(scratch_dir, exist_ok=True)

    output_file_path = os.path.join(scratch_dir, "output.txt")

    # Write Python version and a message to the file
with open(output_file_path, "w") as f:
        f.write(f"Hello from scratch I/O!\n")
        f.write(f"Python Version: {sys.version}\n")

    print(f"Output written to {output_file_path}")

if __name__ == '__main__':
    main()
EOF

cat scratch_io_script.py
```

```
import sys import
os

def main():     if
len(sys.argv) != 2:
print("Usage: python
scratch_io_script.py
<scratch_directory_path>")
sys.exit(1)
```

```
    scratch_dir = sys.argv[1]

    # Create the scratch directory if it doesn't exist
os.makedirs(scratch_dir, exist_ok=True)

    output_file_path = os.path.join(scratch_dir, "output.txt")

    # Write Python version and a message to the file
with open(output_file_path, "w") as f:
        f.write(f"Hello from scratch I/O!\n")
        f.write(f"Python Version: {sys.version}\n")

    print(f"Output written to {output_file_path}")

if __name__ == '__main__':
    main()
```

```
%%bash

# 1. Create a Python virtual environment named 'my_venv'
echo "Creating virtual environment 'my_venv'..." python3
-m venv my_venv

# 2. Activate the 'my_venv' virtual environment
# Note: Activation typically modifies the shell's PATH and prompts. # In a
single bash cell, we can source it and then run commands within the same
shell context.
echo "Activating virtual environment 'my_venv'..." source
my_venv/bin/activate

# 3. Install a dummy package (e.g., requests) into the activated virtual
environment
echo "Installing 'requests' package..." pip
install requests

# 4. Verify that the virtual environment is active and the package is
available
echo "Verifying virtual environment and package installation..." python -c
"import sys; print(f'Python executable: {sys.executable}'); import requests;
print('requests package successfully imported.')"

# Deactivate the virtual environment to clean up for subsequent steps if
needed deactivate
echo "Virtual environment deactivated."
Creating virtual environment 'my_venv'...
Activating virtual environment 'my_venv'...
```

```
Installing 'requests' package...
Requirement already satisfied: requests in
/usr/local/lib/python3.12/distpackages (2.32.4)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.12/dist-packages (from requests) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests) (2025.11.12)
Verifying virtual environment and package installation... Python
executable: /usr/bin/python3 requests package successfully imported.
Virtual environment deactivated.

Error: Command '['/content/my_venv/bin/python3', '-m', 'ensurepip', '-
upgrade', '--default-pip']' returned non-zero exit status 1. bash: line
10: my_venv/bin/activate: No such file or directory bash: line 21:
deactivate: command not found

%%bash set -e # Exit immediately if a command exits with a non-zero
status.

# Clean up any previous attempts
echo "Cleaning up 'my_venv' directory if it exists..." rm
-rf my_venv

# 1. Create a Python virtual environment named 'my_venv' without installing
pip initially
echo "Creating virtual environment 'my_venv' without default pip..." python3
-m venv --without-pip my_venv

# 2. Activate the 'my_venv' virtual environment echo
"Activating virtual environment 'my_venv'..."
. my_venv/bin/activate # Use '.' for sourcing, which is POSIX compliant

# 3. Manually install pip into the virtual environment echo
"Installing pip into 'my_venv' using get-pip.py..." curl -
sS https://bootstrap.pypa.io/get-pip.py | python

# 4. Install a dummy package (e.g., requests) into the activated virtual
environment
echo "Installing 'requests' package..." pip
install requests

# 5. Verify that the virtual environment is active and the package is
available
```

```
echo "Verifying virtual environment and package installation..." python -c
"import sys; print(f'Python executable: {sys.executable}'); import requests;
print('requests package successfully imported.')"

# Deactivate the virtual environment to clean up for subsequent steps if
needed deactivate echo "Virtual environment deactivated."

Cleaning up 'my_venv' directory if it exists...
Creating virtual environment 'my_venv' without default pip...
Activating virtual environment 'my_venv'...
Installing pip into 'my_venv' using get-pip.py...
Collecting pip
  Downloading pip-25.3-py3-none-any.whl.metadata (4.7 kB)
Downloading pip-25.3-py3-none-any.whl (1.8 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.8/1.8 MB 16.8 MB/s  0:00:00
Installing collected packages: pip
Successfully installed pip-25.3 Installing
'requests' package...
Collecting requests
  Downloading requests-2.32.5-py3-none-any.whl.metadata (4.9 kB)
Collecting charset_normalizer<4,>=2 (from requests)
Downloading charset_normalizer-3.4.4-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata
(37 kB)
Collecting idna<4,>=2.5 (from requests)
  Downloading idna-3.11-py3-none-any.whl.metadata (8.4 kB)
Collecting urllib3<3,>=1.21.1 (from requests)
  Downloading urllib3-2.6.2-py3-none-any.whl.metadata (6.6 kB)
Collecting certifi>=2017.4.17 (from requests)
  Downloading certifi-2026.1.4-py3-none-any.whl.metadata (2.5 kB)
Downloading requests-2.32.5-py3-none-any.whl (64 kB) Downloading
charset_normalizer-3.4.4-cp312-
cp312manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl
(153 kB)
Downloading idna-3.11-py3-none-any.whl (71 kB)
Downloading urllib3-2.6.2-py3-none-any.whl (131 kB)
Downloading certifi-2026.1.4-py3-none-any.whl (152 kB)
Installing collected packages: urllib3, idna, charset_normalizer, certifi,
requests

Successfully installed certifi-2026.1.4 charset_normalizer-3.4.4 idna-3.11
requests-2.32.5 urllib3-2.6.2
Verifying virtual environment and package installation...
Python executable: /content/my_venv/bin/python requests
package successfully imported. Virtual environment
deactivated.
```

```
%%bash
cat <<EOF > submit_venv_job.sh
#!/bin/bash
#SBATCH --job-name=VenvTest
#SBATCH --output=slurm-venv-%j.out
#SBATCH --error=slurm-venv-%j.err
#SBATCH --ntasks=1
#SBATCH --time=00:05:00

echo "Starting Slurm job with virtual environment..."

# Activate the virtual environment source
my_venv/bin/activate

# Verify Python executable and version within the activated venv python
-c "import sys; print(f'Python executable: {sys.executable}');
print(f'Python version: {sys.version}')"

# Execute the Python script with a simulated scratch directory python
scratch_io_script.py scratch_data

# Deactivate the virtual environment (optional, but good practice)
deactivate

echo "Slurm job with virtual environment finished."
EOF

cat submit_venv_job.sh
```

```
#!/bin/bash
#SBATCH --job-name=VenvTest
#SBATCH --output=slurm-venv-%j.out
#SBATCH --error=slurm-venv-%j.err
#SBATCH --ntasks=1
#SBATCH --time=00:05:00

echo "Starting Slurm job with virtual environment..."

# Activate the virtual environment source
my_venv/bin/activate

# Verify Python executable and version within the activated venv python
-c "import sys; print(f'Python executable: {sys.executable}');
print(f'Python version: {sys.version}')"

# Execute the Python script with a simulated scratch directory python
scratch_io_script.py scratch_data
```

```
# Deactivate the virtual environment (optional, but good practice) deactivate

echo "Slurm job with virtual environment finished."

%%bash

JOB_ID=$(date +%s) # Simulate a unique job ID

# Simulate Slurm execution by running the script and redirecting output
bash submit_venv_job.sh > slurm-venv-${JOB_ID}.out 2> slurm-
venv${JOB_ID}.err

echo "Simulated execution of submit_venv_job.sh. Output in slurm-
venv${JOB_ID}.out, Error in slurm-venv-${JOB_ID}.err"
```

```
Simulated execution of submit_venv_job.sh. Output in slurm-venv-
1767676308.out, Error in slurm-venv-1767676308.err
```

```
%%bash ls -l slurm-venv-*.out slurm-
venv-*.err
```

```
-rw-r--r-- 1 root root   0 Jan  6 05:11 slurm-venv-1767676308.err
-rw-r--r-- 1 root root 248 Jan  6 05:11 slurm-venv-1767676308.out
```

```
%%bash
JOB_ID=$(ls slurm-venv-*.out | head -n 1 | sed 's/slurm-venv-//;s/\.out//')
cat slurm-venv-${JOB_ID}.out cat scratch_data/output.txt
```

```
Starting Slurm job with virtual environment...
Python executable: /content/my_venv/bin/python
Python version: 3.12.12 (main, Oct 10 2025, 08:52:57) [GCC 11.4.0]
Output written to scratch_data/output.txt Slurm job with virtual
environment finished.
Hello from scratch I/O!
Python Version: 3.12.12 (main, Oct 10 2025, 08:52:57) [GCC 11.4.0]
```